

7.10 PreQuiz: Logarithmic and Exponential Functions — Markscheme

1. [Maximum mark: 6]

The graph of $f(x) = \log_3(x + 1) + 2$ is shown.

(a) Vertical asymptote where the argument of the log is zero: $x + 1 = 0$. **A1**

$$x = -1$$

(b) Set $f(x) = 0$: $\log_3(x + 1) + 2 = 0$, so $\log_3(x + 1) = -2$. **M1**

$$x + 1 = 3^{-2} = \frac{1}{9} \quad \textbf{A1}$$

$$x = -\frac{8}{9} \quad \textbf{A1}$$

Accept $x \approx -0.889$ (3 s.f.).

(c) Range: $f(x) \in \mathbb{R}$ (all real numbers). **A1**

(d) $f(8) = \log_3(9) + 2 = 2 + 2 = 4$, so the point is $(8, 4)$. **A1**

2. [Maximum mark: 6]

Let $g(x) = 4 \cdot 2^{x-1} - 3$.

(a) As $x \rightarrow -\infty$, $2^{x-1} \rightarrow 0$, so $g(x) \rightarrow -3$.

Horizontal asymptote: $y = -3$.

A1

(b) $g(3) = 4 \cdot 2^{3-1} - 3 = 4 \cdot 4 - 3$

M1

$= 13$

A1

(c) Let $y = 4 \cdot 2^{x-1} - 3$. Interchange x and y and solve for y :

$$x = 4 \cdot 2^{y-1} - 3$$

$$x + 3 = 4 \cdot 2^{y-1}$$

M1

$$2^{y-1} = \frac{x+3}{4}$$

$$\text{Take } \log_2 \text{ of both sides: } y - 1 = \log_2 \left(\frac{x+3}{4} \right)$$

A1

$$g^{-1}(x) = \log_2 \left(\frac{x+3}{4} \right) + 1$$

A1

Accept equivalent forms, e.g. $g^{-1}(x) = \log_2(x+3) - 1$.

3. [Maximum mark: 6]

$$N(t) = 8000 e^{kt}, \text{ with } N(3) = 24\,000.$$

(a) $8000 e^{3k} = 24\,000$, so $e^{3k} = 3$. **M1**

$3k = \ln 3$, hence $k = \frac{1}{3} \ln 3$ ($a = 3$). **A1**

(b) $N(7) = 8000 e^{7k} = 8000 \cdot 3^{7/3}$.

GDC: $8000 \times e^{(7/3) \ln 3} = 8000 \times 12.9805 \dots$ **M1**

$N(7) \approx 103\,844$ users. **A1**

(c) Set $8000 e^{kt} = 72\,000$, so $e^{kt} = 9$.

$kt = \ln 9 = 2 \ln 3$ **M1**

$t = \frac{2 \ln 3}{\frac{1}{3} \ln 3} = 6$ years. **A1**

Accept $t \approx 6.00$ (3 s.f.) or by GDC solving $8000 e^{(t \ln 3)/3} = 72000$.

4. [Maximum mark: 6]

Sofia invests \$2500 at 4 % per annum, compounded annually.

(a) End-of-year-1 value $= 2500 \times 1.04 = 2600$. **AG**

(b) $V_{10} = 2500 \times 1.04^{10}$ **M1**

$$= 2500 \times 1.48024 \dots$$

$= \$3700.61$ (nearest cent). **A1**

(c) Need the smallest integer n with $2500 \times 1.04^n \geq 5000$, i.e. $1.04^n \geq 2$. **M1**

Take logs: $n \geq \frac{\ln 2}{\ln 1.04} = 17.673 \dots$ **A1**

Smallest full-year $n = 18$ years. **A1**

GDC check: $1.04^{17} = 1.948 \dots$ (*under*), $1.04^{18} = 2.026 \dots$ (*over*); equivalently $2500 \times 1.04^{17} = 4870.38$ and $2500 \times 1.04^{18} = 5065.20$.

5. [Maximum mark: 5]

$$M = \log_{10}\left(\frac{I}{I_0}\right).$$

(a) $5.4 = \log_{10}\left(\frac{I}{I_0}\right)$, so $\frac{I}{I_0} = 10^{5.4}$. **M1**

$= 251\,188.6 \dots = 2.51 \times 10^5$ (3 s.f.). **A1**

(b) $I_B = 500 I_A$, so

$$M_B = \log_{10}\left(\frac{500 I_A}{I_0}\right) = \log_{10}(500) + \log_{10}\left(\frac{I_A}{I_0}\right) \quad \text{M1}$$

$$= \log_{10}(500) + 5.4 \quad \text{A1}$$

$$= 2.69897 \dots + 5.4 = 8.09897 \dots$$

$$M_B \approx 8.10 \text{ (3 s.f.)}. \quad \text{A1}$$

Accept any equivalent method, e.g. $M_B = \log_{10}(500 \times 10^{5.4})$ evaluated by GDC.

6. [Maximum mark: 4]

$$L(d) = L_0 e^{-0.32 d}.$$

$$\text{(a)} \quad \frac{L(4)}{L_0} = e^{-0.32 \times 4} = e^{-1.28}. \quad \text{M1}$$

$$= 0.27804 \dots \approx 27.8 \% \text{ (3 s.f.)}. \quad \text{A1}$$

$$\text{(b)} \quad \text{Set } e^{-0.32 d} = \frac{1}{2}.$$

$$-0.32 d = \ln \frac{1}{2} = -\ln 2 \quad \text{M1}$$

$$d = \frac{\ln 2}{0.32} = 2.1660 \dots \approx 2.17 \text{ m (3 s.f.)}. \quad \text{A1}$$